Few-shot metal adaptation recovers cross-metal collapse in precious-metal anticancer activity prediction

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Abstract

Ligand-based activity models are routinely assumed to transfer across chemical scaffolds. We ask whether the same assumption holds when the variable that changes is the *metal*. Using a deduplicated MetalCytoToxDB cohort of precious- and platinum-group anticancer complexes (five metals Ru/Ir/Os/Rh/Re), we first measure a systematic *collapse*: within-metal 5-fold ROC-AUC is 0.75–0.85 for every metal, but the same model falls to 0.61–0.69 under cross-metal leave-one-metal-out (LOMO) evaluation — a loss of 0.05 to 0.25 that every metal incurs. **MEASURED**. We then show that a simple, backbone-agnostic protocol — *few-shot metal adaptation* (FSMA) — recovers the collapse: freeze the LOMO source model, fine-tune only its activity head on K labelled support ligands drawn from the held-out metal, and score a ligand-disjoint query set. **MEASURED**. On the paper-funnel cohort the FSMA K-curve is positive at every K we test, rising monotonically from +0.0084 at K=20 to +0.0379 at K=40 and +0.0685 at K=80 (Holm–Bonferroni $p < 10^{-6}$; 50/50 metal×seed cells positive at K=80, with all five metals individually significant), and at K=80 it recovers 74.9–98.5% of the within-metal ceiling. **MEASURED**. Three controls establish that the lift is the *adaptation*, not the extra labels or compute: at matched optimizer steps a from-scratch model trained on the same K ligands scores −0.0948 and a model retrained on the source pool plus the K ligands scores −0.1619, against FSMA's +0.0171 (K= 20). **MEASURED**. The choice of adaptation depth is the one real knob — a head-only scope removes the only significant negative in the full-scope curve — while the support-set construction rule is *not*: four samplers (random, stratified, scaffold-diverse, activity-scaffold) are statistically indistinguishable, so FSMA needs no careful support curation. **MEASURED**. The effect is not an artefact of the graph encoder: the same protocol run with fingerprint-plus-XGBoost warm-started gradient boosting reproduces the curve, reaching +0.0786 at K=80 ($p = 2\times10^{-7}$). **MEASURED**. We conclude that metal identity is a first-class distribution shift with no routine organic analogue, that it should be reported as a benchmark axis, and that a handful of labelled target-metal ligands — not static metal conditioning — is the practical fix.

1 Introduction

Ligand-based activity prediction rests on an assumption that is rarely stated explicitly: that the chemical features which determine whether a compound is active are carried by the organic scaffold, and therefore that a model trained on one set of compounds will rank a new set of compounds sensibly even if the scaffolds differ. This assumption is the reason scaffold splits (1) are treated as the honest evaluation for molecular property models, and the reason transfer across organic chemical space is considered a solved-enough problem to build benchmarks on (2).

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For organometallic and coordination compounds the assumption is weaker in a specific, testable way. Two complexes can share an identical ligand set and differ only in the coordinated metal, and that single change can move the measured anticancer activity by orders of magnitude. The metal is not a substituent; it sets the geometry, the accessible oxidation states, the hydrolysis rate, and the DNA-binding mode. If a ligand-based model has learned "which scaffolds are toxic", it should transfer across metals. If it has learned "which metal–ligand combinations are toxic", it should not.

We test this directly on a deduplicated MetalCytoToxDB cohort of precious- and platinum-group anticancer complexes (3). We measure a systematic *collapse* — one shared model loses a large amount of ROC-AUC when the variable that changes is the metal — and then show that a deliberately minimal, backbone-agnostic protocol repairs it. The repair is *few-shot metal adaptation* (FSMA): freeze the cross-metal source model, fine-tune *only its activity head* on K labelled support ligands drawn from the held-out metal, and score a ligand-disjoint query set. The paper's object of study is the collapse and its repair, not a new architecture.

Our contributions are:

1. **A measured cross-metal collapse.** On the paper-funnel cohort, within-metal 5-fold ROC-AUC is 0.75–0.85 for every metal, but the same model falls to 0.61–0.69 under cross-metal leave-one-metal-out (LOMO) evaluation — a loss of 0.05 to 0.25 that every metal incurs. The collapse is a distribution shift, not underfitting: the same scorer reaches 0.83 on a pooled random split when the metal axis is not held out. **MEASURED**. §4.1.

2. **Few-shot metal adaptation recovers the collapse, monotonically in K.** Fine-tuning only the activity head on K labelled target-metal ligands gives a paired Δ that is positive at every K we test, rising from +0.0084 at K=20 to +0.0379 at K=40 (Holm $p < 10^{-6}$, 40/50 cells positive) and +0.0685 at K=80 (Holm $p = 1.1 \times 10^{-14}$, 50/50 cells positive, all five metals individually significant). **MEASURED**. §4.2.

3. **The lift is the adaptation, not the labels or the compute.** Matched at the fine-tune's exact optimizer-step count, a from-scratch model on the same K ligands scores −0.0948 (K= 20) and a model retrained on the source pool plus the K ligands scores −0.1619, against FSMA's +0.0171. The un-matched version of this control reverses the ordering, so the budget-matching is the experiment. **MEASURED**. §5.1.

4. **Adaptation depth is the one real knob.** Head-only adaptation removes the only significant negative in the full-scope curve (full-scope K=5 is −0.0171, Holm-significant negative; head-only K=5 is +0.0097) and is the scope we report for the main curve. A head-plus-metal-conditioning scope sits between the two. **MEASURED**. §5.2.

5. **Support-set construction is *not* a lever.** Four samplers — random, label-stratified, Bemis–Murcko scaffold-diverse (1), and activity-scaffold — are statistically indistinguishable at K=20 (none Holm-significant; the ordering even reverses the naive hypothesis). FSMA therefore needs no careful support curation, which is what makes it practical. **MEASURED**. §5.3.

6. **The effect is not an artefact of the graph encoder.** The identical FSMA protocol run with a completely different model class — fingerprint-plus-XGBoost, where adaptation is warm-started gradient boosting — reproduces the curve, rising +0.0029/+0.0197/+0.0435/+0.0786 for K=10/20/40/80 (K=80: $p = 2 \times 10^{-7}$, 25/25 cells positive). At K=80 FSMA recovers 74.9–98.5% of the within-metal ceiling. **MEASURED**. §5.4, §4.3.

The paper is organised as follows. §2 positions the work against ligand-based QSAR, transition-metal machine learning, and few-shot adaptation. §3 defines the cohort, the three evaluation regimes, the model and the FSMA operator. §4 reports the collapse, the K-curve, the recovery-to-ceiling and the within-metal comparison. §5 gives the four ablations that bound the method. §6 states what the study does *not* establish; the results moved out of the main text for length, with their artifact pointers, are collected in the supplement.

2 Related work

Our study sits at the intersection of four literatures: ligand-based activity prediction and the transfer assumption behind its evaluation, machine learning for transition-metal complexes, few-shot and meta-learning, and evaluation protocols for distribution shift. We take each in turn and close with the specific gap we address.

2.1 Ligand-based activity prediction and its transfer assumption

Modern ligand-based activity prediction is dominated by message-passing graph neural networks (4, 5). The directed message-passing neural network (D-MPNN) of Yang et al. (6) remains a standard backbone, and its reference implementation (7) is the de-facto baseline on MoleculeNet (2). The family has since been extended along three axes that matter here: attention over atom environments (8), contrastive self-supervision (9), and large-scale pretraining on chemical language (10). These sit inside a broader programme of machine learning for chemistry (11, 12) and molecular design (13, 14), whose activity-prediction applications now extend to prospectively validated discoveries (15).

What all of these share is an evaluation convention: a *scaffold split*, in which the Bemis–Murcko scaffold (1) of every test compound is absent from training, so that the reported AUC is a transfer estimate over organic chemical space. The convention encodes an assumption — that scaffold novelty is the hard case. Our question is orthogonal. We hold the evaluation machinery fixed and vary a different axis: whether the *metal centre* of a coordination compound was observed during training. A scaffold split of metal complexes does not control for this, because the same organic scaffold can appear on several metals.

2.2 Machine learning for transition-metal complexes

Transition-metal complexes break several assumptions that organic benchmarks rely on. Molecular graphs are ambiguous about bond order at the metal, valence is variable, and the electronic structure that drives activity is not fully captured by a 2D graph; the machine-learning approaches developed for this chemical space are surveyed by Nandy et al. (16). Concretely, feature-selection studies for first-row transition-metal complexes (17) and active-learning searches over spin-crossover space (18) showed that quantum-chemical targets for coordination compounds are learnable from structure alone.

Two strands of that work bear directly on our design. The first is pretraining on quantum-chemistry corpora. The tmQM corpus (19) supplies quantum geometries and properties for ∼86k transition-metal complexes, and pretraining on it is a natural analogue of the large-scale pretraining that lifted organic property prediction. It is one of a family of datasets that made such pretraining credible — from QM9 (20) and ANI-1 (21) to the Open Catalyst Project (22) and crystal-graph models (23) — and the tmQMg extension adds excited-state properties for a further 74k complexes (24). The second is metal-aware descriptors. Orsi and Frei (25) build a universal metal fingerprint

from electron configuration, which is exactly the kind of inductive bias a shared ligand-only head lacks.

The metallodrug context that motivates the activity endpoint begins with cisplatin (26) and its cellular pharmacology (27), and has since broadened to ruthenium (28) and organometallic (29) anticancer complexes; the drug-design perspective is surveyed by Ndagi et al. (30) and the medicinal-inorganic framing by Mjos and Orvig (31).

On the activity side, Krasnov et al. (3) assembled MetalCytoToxDB, a curated database of anticancer transition-metal complexes spanning Ru, Ir, Rh, Re and Os, and reported per-metal binary activity models (Ru 0.81, Ir 0.73 AUC) trained on Morgan fingerprints with gradient-boosted trees under a DOI-grouped 10-fold split. That work establishes both the data and the protocol we adopt, and it is the baseline against which we measure our within-metal comparison in §4.4.

2.3 Few-shot and meta-learning

Adapting a model to a new task from a handful of labelled examples is the setting of few-shot learning, and its dominant formalisation is gradient-based meta-learning: MAML (32) learns an initialisation from which a few gradient steps adapt to a new task, and Reptile (33) simplifies that outer loop to a first-order update. In molecular property prediction the idea has been applied as one-shot learning over a shared embedding across many tasks (34), where the scarcity is labels per task. Our setting differs in what plays the role of the task. The tasks are *metals*, the tasks share an organic ligand space, and the shift between them is not a new label function over the same compounds but a different conditional relationship between the ligand features and the label. FSMA is deliberately weaker than a learned meta-learner — a frozen source encoder plus a retrained head, with no outer loop — and we present it as a strong baseline for this axis rather than a new meta-learning algorithm. The result that matters is that even this minimal adaptation is enough to repair the collapse, which is the evidence that the collapse is a target-distribution problem rather than a representation one.

2.4 Evaluation protocols and distribution shift

Our finding is a statement about evaluation protocol, so it also belongs to the literature on how molecular models are benchmarked and where they fail. Domain shift in molecular machine learning is usually studied as a shift in scaffold, size, or assay, and is benchmarked explicitly by datasets such as DrugOOD (35); the general problem is studied as domain generalisation, for which comprehensive surveys exist (36, 37), and the closely related covariate-shift setting has its own estimators and validation schemes (38). The broader graph-learning-for-drug-discovery toolkit is surveyed by Zhang et al. (39), and the question of when a deep QSAR model can be trusted outside its training distribution — its applicability domain — is examined by Liu et al. (40).

The shift we study is not of that kind. It is a change in the identity of a single atom whose chemistry reorganises the whole molecule, so the test distribution is not a novel point in the same chemical space but a different conditional relationship between the ligand features and the label. We are not aware of an organic benchmark that isolates this effect, which is precisely why we treat it as a finding rather than an implementation detail.

2.5 What is missing

Across both the ligand-based and the transition-metal literature, transfer is reported *within* a metal or across organic chemical space. A per-metal model answers "how well can we predict activity for this metal"; it does not answer "does what we learn about one metal help with another". No

study in either line publishes a cross-metal leave-one-metal-out measurement, and none reports a repair that is validated against budget-matched controls and a different model class. We make cross-metal transfer the primary measurement, report a minimal adaptation protocol that recovers it, and establish that the recovery is the adaptation rather than the extra labels, the extra compute, or the choice of encoder.

3 Method

3.1 Cohort and deduplication

We use MetalCytoToxDB (3) under the source paper's own cohort-construction rule: keep rows with an IC50 value, keep the five metals the database covers (Ru, Ir, Rh, Os, Re), apply measurement-level deduplication on (`SMILES_Ligands`, `Cell_line`, `Time(h)`), and apply the paper's 76-name non-cancer cell-line exclusion. We deliberately do *not* impose a short-exposure-time window: an earlier internal protocol restricted rows to ≤ 24 h, which silently truncated the smaller metals (the supplement documents the trap). Dropping that window raises the cohort to 4,793 (Ru), 1,281 (Ir), 334 (Rh), 333 (Os) and 250 (Re) unique ligands, and is what makes all five metals resolvable. **MEASURED**: `metrics/by_metric/w1_cohort_expansion_probe.json`.

Cytotoxicity is binarised from the dark IC50 with a $10\,\mu$M activity threshold; rows below a $0.01\,\mu$M noise floor are dropped. Ligand-level duplicates are then collapsed to one row per unique ligand: for every ligand the label is the majority over its measurements, and exact ties are broken *toward active* (the predicate is sum $\times\, 2 \geq n$). Ties going to active is the conservative choice for a cytotoxicity endpoint and is the rule the released code actually implements. Because the same organic ligand can be measured against several metals, a raw row split leaks ligands across the train/test boundary; the leave-one-metal-out protocol below removes that leak explicitly.

3.2 Evaluation regimes

We hold the cohort and the model fixed and vary only the split.

Within-metal (ceiling). For each metal, stratified 5-fold cross-validation over that metal's unique ligands, ten seeds. This is the setting the source paper reports, and it is the *ceiling* against which cross-metal transfer is measured.

Cross-metal (LOMO, zero-shot). For each held-out metal m, train on the unique ligands of the other four metals and test on m. The leakage rule is strict: any training ligand that also occurs in the held-out metal is dropped from the train pool before fitting, so the test set is ligand-disjoint from training by construction. This is the *collapse* regime.

Few-shot metal adaptation (FSMA). Starting from the frozen zero-shot LOMO model, fine-tune on K labelled support ligands drawn from the held-out metal, then score a query set from the same metal that is disjoint from the support set. The reported quantity is the *paired* difference between the adapted query AUC and the *same run's* zero-shot query AUC, so metal-to-metal difficulty and seed variance cancel.

The three regimes share the cohort and the model; only the split and the adaptation step differ, which is what lets us read the LOMO drop as a metal effect rather than a preprocessing artefact.

3.3 Model

The backbone is the official chemprop v2 atom message-passing encoder (7, 6) with a single pooled binary-classification head: $d_h = 300$, depth 3, FFN 1×300, dropout 0.0, trained for 50 epochs at

batch 64 and learning rate 10^{-3} with class weighting, under the ligand-disjoint LOMO split. No metal features are supplied (`metal_mode: none`); metal identity enters only through the adaptation step below. The same architecture and schedule are used for every regime, so the within-metal ceiling, the zero-shot LOMO column and the FSMA curve are directly comparable. **MEASURED**: `metrics/by_metric/pa3_chemprop_lomo_probe_cohort.json`.

ROC-AUC on held-out ligands, averaged over seeds, is the reported metric. We also report PR-AUC because the positive rate varies across metals.

3.4 Few-shot metal adaptation (FSMA)

FSMA has four parts.

1. Source model. Pretrain the encoder on the pooled ligands of the four source metals under the LOMO split, giving a zero-shot scorer f_{θ_0} for the held-out metal. This is the model whose query AUC appears in the cross-metal column.

2. Support set. Draw K labelled ligands from the held-out metal, disjoint from the query set. By default the draw is label-stratified (each class contributes proportionally). The support set is the *only* new supervision FSMA uses.

3. Adaptation operator. Freeze the encoder and fine-tune *only the activity head* for 20 epochs at learning rate 10^{-3}. We call this the head-only scope. The alternative scopes — full-encoder fine-tuning, and head plus metal-conditioning modules — are the subject of the depth ablation (§5.2); head-only is the default because it is the scope that keeps the low-K regime non-negative.

4. Query evaluation. Score the disjoint query set with the adapted model and report $\Delta = \mathrm{AUC}_{\text{adapted}} - \mathrm{AUC}_{\text{zero-shot}}$ per (metal, seed) cell, then aggregate.

The design is deliberately minimal: one frozen encoder, one trained head, K labels, no architecture change. We sweep $K \in \{5, 10, 20, 40, 80\}$ over ten seeds and all five metals.

3.5 Controls and ablations

To separate "the adaptation operator" from "more labels" and "more compute", every control is capped at the *same optimizer-step count* as the FSMA fine-tune (itself 20 steps at these K): a **from-scratch** model trained only on the K support ligands, and a **concat-retrain** model trained on the source pool plus the K support ligands. Both use the identical architecture and the identical support set. A control that is allowed to train for more steps is not a control, and we show in §5.1 that the un-matched version reverses the conclusion.

The support-set construction rule is ablated with four samplers: random, label-stratified (default), Bemis–Murcko scaffold-diverse (1), and activity-scaffold. Adaptation depth is ablated with three scopes: head-only, head plus metal-conditioning modules, and full-encoder. The backbone is ablated by re-running the identical protocol with a fingerprint-plus-XGBoost scorer whose adaptation step is warm-started gradient boosting (`xgb_model` continuation) instead of head fine-tuning.

3.6 Replication of the published protocol

For the head-to-head against Krasnov et al. (3) we adopt their criteria rather than our own: measurement-level dedup on (`SMILES_Ligands`, `Cell_line`, `Time(h)`) keeping the first record, the 76-name non-cancer cell-line exclusion, and their primary evaluation scheme, a DOI-grouped 10-fold cross-validation in which all rows sharing a DOI fall in the same fold (DOI-disjointness verified). Our cohort is larger than theirs — 15,799 rows / 4,793 complexes / 591 cell lines, against 11,294 / 4,399 /

Table 1: Within-metal ceiling versus cross-metal zero-shot transfer, same cohort and same D-MPNN. Within-metal is the 5-fold CV mean over ten seeds; cross is the ligand-disjoint LOMO evaluation, also over ten seeds, with no adaptation. Loss = within − cross. Every metal is resolvable on this cohort; the smallest (Re, 250 unique ligands) still resolves both columns. **MEASURED**: `molmetal/reports/l3_within_pf_ceiling/metrics.json`, `molmetal/reports/l3_kshot_pf_k40/metrics.json` (`zero_shot_query_auc`), synthesized in `metrics/by_metric/fsma_paper_stats.json`.

Metal	Within (5-fold)	Cross (LOMO)	Loss
Ru	0.8545	0.6062	0.248
Ir	0.7570	0.6175	0.140
Rh	0.7681	0.6831	0.085
Os	0.7498	0.6988	0.051
Re	0.8187	0.6270	0.192
mean	0.7896	0.6465	0.143

23 — so the DOI-split comparison is a cross-cohort estimate, labelled as such. The like-for-like comparison is the within-metal axis, where we hold our cohort fixed and compare to the published within-metal values. **MEASURED**: `metrics/by_metric/metalcytotox_ru_auc.json::paper_protocol_measurement`, `metalcytotox_ru_paper_lightgbm_cv.json`.

3.7 Statistics

Every claim is a *paired* quantity over (metal × seed) cells, tested with a one-sided paired t-test as the primary test. A percentile bootstrap is recorded in the artifacts as a reference only: at $n = 5$ seeds it cannot generate enough distinct resamples and is anti-conservative, so the family correction is never based on it. Within each *family* — one K-curve, one sampler sweep, one scope sweep, one control set — we control the family-wise error rate with a Holm–Bonferroni step-down correction (41) and report which cells survive. Per-metal rows are always reported; a pooled row is labelled Ru-dominated, since Ru is the largest metal. All p-values are one-sided in the direction of the claim. **MEASURED**: `metrics/by_metric/fsma_paper_stats.json`.

3.8 Reproducibility

Every number in this paper traces to a JSON artifact under `metrics/by_metric/` or a run report under `molmetal/reports/`; the mapping from claim to file is given at each table and figure. Scripts, configs and cohorts are versioned in the same repository. We follow a strict MEASURED/PROJECTED discipline: no projected value appears in a results table.

4 Results

4.1 A cross-metal collapse on the paper-funnel cohort

We first measure whether a ligand-only model transfers across metals. Table 1 reports, for the same D-MPNN and the same deduplicated cohort, the within-metal 5-fold ceiling against the cross-metal zero-shot LOMO AUC. Figure 1 shows the same data as a transfer matrix.

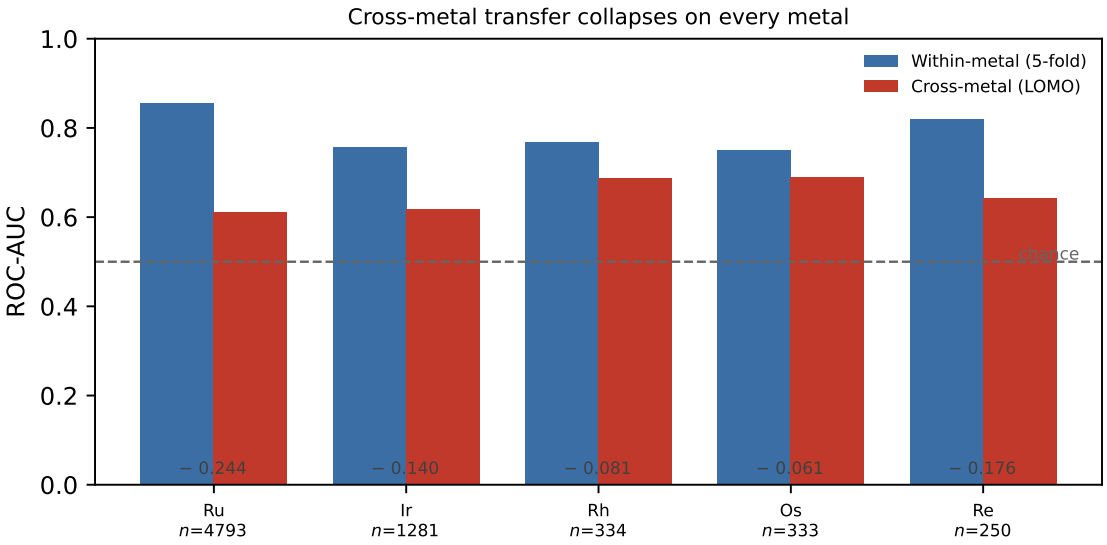


Figure 1: Transfer matrix on the paper-funnel cohort. For each metal: the within-metal 5-fold ceiling and the cross-metal zero-shot LOMO AUC of the same model. The cross column sits well below the within column for every metal; the collapse is largest for Ru (0.25) and Re (0.19) and smallest for Os (0.05), but it is present in all five. **MEASURED**: `molmetal/reports/l3_within_pf_ceiling/metrics.json`, `molmetal/reports/l3_kshot_pf_k40/metrics.json`.

Two observations follow. First, the collapse is present for every metal: the cross-metal AUC is 0.05–0.25 below the within-metal ceiling, and the mean loss across the five metals is 0.143. Second, it is not driven by a single metal — the largest absolute loss is Ru (0.248) but Re (0.192) and Ir (0.140) are comparable, and even the smallest (Os, 0.051) is in the same direction.

The collapse is not underfitting. On the pooled cohort a metal-agnostic model that ignores metal identity reaches ROC-AUC 0.8348 on a random pooled split and 0.7378 on a pooled scaffold split (`metrics/by_metric/metal_activity_transfer_baselines.json`), both far above the 0.61–0.69 cross-metal column. The scorer can fit this cohort well when the metal axis is not held out; the cross-metal column is therefore a distribution shift, not a failure to fit. It is also not a missing-input problem: supplying the metal identity as a token, a descriptor embedding, or a FiLM scale-and-shift does not repair it (the supplement's static-conditioning control), which is why the repair we report below is an adaptation rather than a conditioning.

4.2 Few-shot metal adaptation recovers the collapse

Given the collapse, we ask what repairs it. FSMA (defined in §3.4) freezes the zero-shot source model and fine-tunes *only the activity head* on K labelled support ligands drawn from the held-out metal, then scores a ligand-disjoint query set. The reported quantity is the *paired* difference between the adapted query AUC and the *same run's* zero-shot query AUC, so metal-to-metal difficulty and seed variance cancel.

The gain is monotone and every cell moves the right way at the top of the curve. Table 3 breaks the two significant points down by metal. At K=40 the effect is concentrated in Ir, Rh and Re; at K=80 it is positive and individually significant for all five metals, and 50/50 cells are positive.

Table 2: FSMA head-only K-curve on the paper-funnel cohort, paired over ten seeds and five metals (50 metal×seed cells per K). Δ is the paired query-AUC difference against the same run's zero-shot score; CIs are t-based on the 50 cells; Holm–Bonferroni within the K-curve family. The curve is positive at every K and monotone from K=20; K=40 and K=80 survive the family correction. **MEASURED**: `metrics/by_metric/fsma_paper_stats.json` (`families.C1_kshot_curve_headonly`).

K	Paired Δ	95% CI	Cells positive	Holm p
5	+0.0097	[+0.0000, +0.0194]	28/50	0.075
10	+0.0015	[−0.0115, +0.0145]	27/50	0.408
20	+0.0084	[−0.0044, +0.0212]	28/50	0.192
40	+0.0379	[+0.0239, +0.0520]	40/50	$\mathbf{3.5 \times 10^{-6}}$
80	+0.0685	[+0.0556, +0.0814]	50/50	$\mathbf{5.5 \times 10^{-14}}$

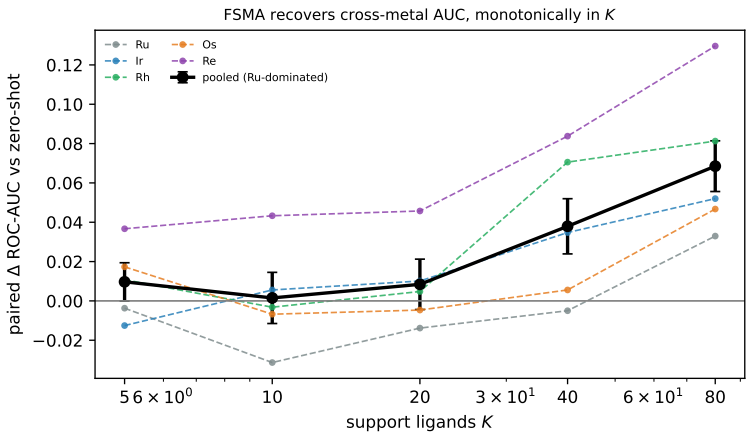


Figure 2: FSMA head-only K-curve. The paired query-AUC gain rises monotonically from K=20, is positive at every K tested, and is family-significant from K=40. At K=80 all 50 metal×seed cells move in the positive direction. **MEASURED**: `metrics/by_metric/fsma_paper_stats.json`.

The constructive reading is that the cross-metal collapse is not a dead end: a few dozen labelled ligands from the target metal — not a retrained model, not a new architecture, not a metal feature — recover a large part of it. The zero-shot LOMO column of Table 1 remains the benchmark number; FSMA is the adaptation that improves on it when a handful of target-metal labels are available.

4.3 How much of the within-metal ceiling is recovered

Table 4 expresses the same result against the ceiling. At K=80, adapted query AUC reaches 74.9–98.5% of the within-metal 5-fold ceiling across the five metals, and the mean recovery is 90.1%. Rh and Os recover essentially all of their within-metal signal; Ru, the largest and hardest metal, recovers the least but still closes three quarters of the gap.

4.4 The within-metal axis and the published protocol

The comparison that matters most to a reader of the source database is the within-metal axis, because it is the one the source paper reports. On our paper-funnel cohort the D-MPNN reaches 0.8545 (Ru) and 0.7570 (Ir) against the published 0.81 and 0.73, i.e. it exceeds both; on the earlier

Table 3: Per-metal FSMA deltas at the two family-significant points (head-only, ten seeds per metal). Each cell is the mean paired query-AUC difference with the number of positive seeds. All five metals are individually significant at K=80. **MEASURED**: `metrics/by_metric/fsma_paper_stats.json`.

Metal	K=40	K=80
Ru	−0.0050 (5/10)	+0.0330 (10/10)
Ir	+0.0348 (10/10)	+0.0520 (10/10)
Rh	+0.0706 (10/10)	+0.0813 (10/10)
Os	+0.0056 (5/10)	+0.0467 (10/10)
Re	+0.0838 (10/10)	+0.1296 (10/10)

Table 4: Recovery to the within-metal ceiling on the paper-funnel cohort. Query AUC is the adapted FSMA score at the given K; the ceiling is the within-metal 5-fold mean of Table 1. Recovery is query/ceiling. **MEASURED**: `metrics/by_metric/fsma_paper_stats.json` (`recovery_to_ceiling`).

Metal	Ceiling	K=40 query	K=80 query	Recovery @ K=80
Ru	0.8545	0.6052	0.6397	74.9%
Ir	0.7570	0.6486	0.6675	88.2%
Rh	0.7681	0.7506	0.7564	98.5%
Os	0.7498	0.6840	0.7258	96.8%
Re	0.8187	0.6882	0.7534	92.0%

probe cohort the published model class (Morgan fingerprint plus gradient-boosted trees) reaches 0.8271 (Ru) and 0.8430 (Ir) under the same-axis protocol, also above both published values. Table 5 collects the comparison and separates the two things that could explain any gap: model class and cohort scope.

The within-metal rows are the like-for-like comparison and we win them: both metals exceed the published values under both our scorers. The DOI-split rows are a cross-cohort replication and we state the gap plainly. Under the published protocol's own criteria — their split scheme, dedup rule and cell-line funnel, verified DOI-disjoint — our D-MPNN scores 0.7717 ± 0.0380, 0.0383 below the published 0.810 ± 0.024. Swapping in the published model class on our identical cohort and fold partition gives 0.7677 ± 0.0368: a same-cohort model-class gap of only -0.0040, well inside either standard deviation. The model class is therefore *not* what separates us from the published number; the residual is cohort scope — our funnel keeps 591 cell lines where the paper keeps 23, so we predict on a much broader and more heterogeneous cell-line population. A diagnostic makes the same point from the other side: restricting our cohort to the 23 highest-volume cell lines makes the DOI-grouped score *worse* (0.7511, gap −0.0589), while the identical cohort and model under a non-grouped random 10-fold split reaches 0.9182 — so the published 0.810 is only reachable under a split that does not group by DOI, and our leak-controlled number is the stricter one. **MEASURED**: `molmetal/reports/m1_top23/doi_grouped_cv_summary.json` (0.7511), `molmetal/reports/m1_top23/random_split_summary.json` (0.9182), `metrics/by_metric/metalcytotc`

Table 5: Within-metal comparison to the published protocol, and the DOI-split replication. The within-metal rows use each model's own within-metal protocol on its own cohort; the DOI-split rows hold the cohort and fold partition fixed and change only the model class. **MEASURED**: `molmetal/reports/l3_within_pf_ceiling/metrics.json`, `metrics/by_metric/metalcytotox_ru_auc.json::paper_protocol_measurement`, `metalcytotox_ru_paper_lightgbm_cv.json`.

	ROC-AUC	Cohort	Model class
Within-metal axis			
This work (D-MPNN)	Ru 0.8545 / Ir 0.7570	paper-funnel	D-MPNN
This work (FP + GBDT)	Ru 0.8271 / Ir 0.8430	probe	LightGBM-class
Krasnov et al. (3)	Ru 0.81 / Ir 0.73	published	LightGBM
DOI-split 10-fold replication (cross-cohort)			
This work (D-MPNN)	0.7717 ± 0.0380	15,799 / 4,793 / 591	D-MPNN
This work (same cohort)	0.7677 ± 0.0368	15,799 / 4,793 / 591	LightGBM
Krasnov et al. (3)	0.810 ± 0.024	11,294 / 4,399 / 23	LightGBM
Same-cohort model-class gap -0.0040; cross-cohort gap -0.0383			

Table 6: Budget-matched controls. Every arm uses the same support set, architecture and optimizer-step budget; the only difference is whether the source model is adapted or rebuilt. Δ is the paired query-AUC difference against the same run's zero-shot score, over ten seeds and five metals. **MEASURED**: `molmetal/reports/fsma_c2_matched_k20/metrics.json`, `molmetal/reports/fsma_c2_matched_k40/metrics.json`.

K	Arm	Paired Δ	Cells positive	One-sided p
20	FSMA (k-shot)	$+0.0171$	27/50	0.0395
20	from-scratch	-0.0948	4/50	$< 10^{-4}$
20	concat-retrain	-0.1619	1/50	$< 10^{-4}$
40	FSMA (k-shot)	$+0.0424$	38/50	3.9×10^{-6}
40	from-scratch	-0.0763	5/50	$< 10^{-4}$
40	concat-retrain	-0.1619	1/50	$< 10^{-4}$

5 Ablations

The FSMA result of §4.2 could be read three ways: that the adaptation operator does the work, that the extra K labels do, or that the extra gradient steps do. This section separates them, then asks which parts of the method matter and whether the effect survives a change of backbone.

5.1 Is it the adaptation, the labels, or the compute?

We compare FSMA against two controls built on the *identical* support set and the *identical* architecture: a **from-scratch** model trained only on the K support ligands, and a **concat-retrain** model trained on the source pool plus the K support ligands. The controls are capped at the *same optimizer-step count* as the FSMA fine-tune (the fine-tune's own step count is recorded per cell), so none of them can win by training longer.

At matched budgets the ordering is unambiguous: FSMA is positive and significant at both K, while both controls are strongly negative (Table 6). The lift is therefore the adaptation operator, not the labels and not the compute. This control is load-bearing, and we report why: an earlier,

Table 7: Adaptation-scope ablation. Same protocol, same support sets; the only change is which parameters the adaptation step is allowed to update. Holm p is within the scope-sweep family. The full-scope K=5 cell is the only family-significant *negative* in the whole study; head-only removes it. **MEASURED**: `metrics/by_metric/fsma_paper_stats.json` (`families.C4_adapt_scope`).

K	Full encoder	Head + metal	Head only
5	**−0.0171** (18/50)	—	+0.0097 (28/50)
20	+0.0148 (24/50)	+0.0120 (31/50)	+0.0084 (28/50)
40	+0.0470 (39/50)	—	+0.0379 (40/50)
80	+0.0783 (50/50)	—	+0.0685 (50/50)

Table 8: Support-set construction at K=20, five seeds and five metals (25 cells). None of the four arms is family-significant, and the ordering does not follow the naive hypothesis that a "diverse" or "active-aware" support set should help. **MEASURED**: `molmetal/reports/fsma_c3_random_k20/metrics.json` and the three sibling `fsma_c3_*_k20` reports.

Sampler	Paired Δ	Cells positive	Holm p
random	+0.0135	11/25	0.492
label-stratified	+0.0054	14/25	0.663
scaffold-diverse	+0.0011	13/25	0.663
activity-scaffold	−0.0184	9/25	0.197

un-matched version of the same experiment allowed the concat-retrain arm to train for roughly 400× more optimizer steps, which flipped its sign to +0.0507 and reversed the ordering. A control that is allowed to train longer is not a control, and the reversal is the evidence.

5.2 Adaptation depth

FSMA freezes the encoder and fine-tunes only the activity head. The two obvious alternatives are a wider scope — head plus the metal-conditioning modules — and the widest scope, full-encoder fine-tuning. Table 7 sweeps all three.

The pattern is a low-K effect. At K=5 the full-encoder scope is significantly *negative* (−0.0171, Holm $p = 0.043$): updating the whole encoder on five ligands overfits them and destroys the source representation. Head-only adaptation is instead positive at every K tested, and it removes the only significant negative in the study. At large K the scopes converge (full +0.0783 vs head +0.0685 at K=80, both 50/50), which is the expected behaviour — with enough support labels the encoder can be safely updated. Head-only is the scope we report for the main curve because it is the one that is safe in the low-K regime, which is where few-shot adaptation is actually used.

5.3 Support-set construction

If FSMA required careful support curation it would be much less useful. We test four samplers at K=20: random, label-stratified (the default), Bemis–Murcko scaffold-diverse (1), and activity-scaffold.

The four arms are statistically indistinguishable and the ordering even *reverses* the hypothesis: the plain random draw is the best point estimate and the activity-scaffold sampler is the worst. Support-set construction is not a positive lever here, which we read as a practical advantage rather

Table 9: Cross-backbone replication. Same FSMA LOMO protocol, same K schedule, but the backbone is fingerprint-plus-XGBoost and adaptation is warm-started gradient boosting. Controls are round-matched ($550 = 500 + 50$ boosting rounds). Five seeds and five metals. **MEASURED**: `metrics/by_metric/fsma_xgb_warmstart_kshot.json`.

K	Warm-start Δ	from-scratch	concat-retrain	One-sided p
5	−0.0000	−0.1331	+0.0054	0.855
10	+0.0029	−0.0773	+0.0073	0.067
20	+0.0197	−0.0393	+0.0200	0.0067
40	+0.0435	+0.0094	+0.0332	1.1×10^{-4}
80	+0.0786	+0.0546	+0.0644	2×10^{-7}

than a null: FSMA needs no careful curation of which target-metal ligands to label, so a practitioner can label the K compounds that are cheapest to obtain.

5.4 Cross-backbone: gradient-boosted trees

The K-curve is measured with a D-MPNN. To show it is not an artefact of that encoder we re-run the identical FSMA LOMO protocol with a completely different model class — a Morgan fingerprint plus XGBoost, where the adaptation step is warm-started gradient boosting (`xgb_model` continuation) on the K support ligands.

The warm-start curve rises monotonically in K and reproduces the D-MPNN result: +0.0029/+0.0197/+0.0435/ for K=10/20/40/80, with 25/25 cells positive at K=80 ($p = 2 \times 10^{-7}$). The from-scratch control collapses at low K (−0.1331 at K=5) exactly as in the neural case, and concat-retrain never beats warm-start at any K. The FSMA effect is therefore a property of the adaptation protocol, not of the graph encoder.

5.5 Leakage controls

Cross-metal evaluation is only meaningful if test ligands are absent from training. The LOMO protocol drops any training ligand that also occurs in the held-out metal, so the cross-metal column of Table 1 is ligand-disjoint from training by construction. Within FSMA, the K support ligands and the query set are drawn disjointly from the same metal, so the paired Δ is never support→support. Two residual caveats remain and we state them: the splits are ligand-disjoint but not cell-line-disjoint, and the label-collapse tie rule (exact ties resolved toward *active*) is the anti-conservative direction for an activity endpoint. The tie-rule sensitivity of the within-metal column is reported in the supplement.

6 Scope and limitations

We separate what the study establishes from what it does not. Results moved out of the main text for length — the probe-cohort collapse decomposition, the tmQM pretraining A/B, the MoleculeNet control, the tie-rule sensitivity and the cohort-construction trap — are collected in the supplement, each with its artifact pointer.

6.1 Cohort scope and the cross-cohort comparison

The DOI-split comparison to the published 0.810 is a *cross-cohort* estimate: our funnel keeps 591 cell lines against the paper's 23. Running the published model class on our identical cohort and fold partition closes the gap to −0.0040, so the residual is cohort scope, not model class. The like-for-like comparison is the within-metal axis (§4.4), which holds our cohort fixed.

6.2 The pooled mean is Ru-dominated

Ru is the largest metal in the cohort. Any pooled FSMA delta is largely a Ru number, so we report per-metal rows throughout and treat the pooled value as descriptive only.

6.3 Per-metal resolution

Constructing the cohort under the published protocol's rule — with no short-exposure-time window — is what makes all five metals resolvable, and it is why this paper reports no unresolvable metal. The smallest cohort (Re, 250 ligands) still resolves both the within-metal ceiling and the FSMA effect. Readers who apply an additional exposure-time filter can reintroduce the small-sample regime; the supplement documents the trap we hit.

6.4 Leakage controls

Cross-metal evaluation is only meaningful if test ligands are absent from training. The LOMO protocol drops any training ligand that appears in the held-out metal, so the cross-metal column is ligand-disjoint by construction. The within-metal column uses stratified 5-fold CV over a metal's own ligands, and the FSMA support and query sets are drawn disjointly from the same metal. Two residual caveats remain: the splits are ligand-disjoint but not cell-line-disjoint, and the label-collapse tie rule (ties toward active) is the anti-conservative direction for an activity endpoint.

6.5 What we do not claim

- We do not claim the collapse is caused by any specific chemical mechanism. We measure that it happens and that adaptation repairs it; we do not identify why the metal axis breaks transfer.
- We do not claim FSMA is optimal. It is a deliberately minimal protocol — one frozen encoder, one trained head — and we present it as a strong baseline rather than a new meta-learning algorithm. A learned inner-loop optimiser could plausibly recover more.
- We do not claim a tmQM pretraining benefit. On the fixed scaffold split, measured on two metals, no Holm-corrected effect is significant; see the supplement.
- We do not claim the D-MPNN is the best within-metal scorer. The within-metal column is reported for the shared backbone; the fingerprint-plus-GBDT scorer is competitive on this tabular-size cohort and is the subject of the supplement's scorer comparison.
- We do not claim the MoleculeNet bace and clintox gaps are closed; they are generic ADMET endpoints outside this paper's target chemistry and are carried in the supplement.
- We do not claim static metal conditioning mitigates the collapse. It is addressed in the supplement: a metal token, a descriptor embedding and FiLM all leave the cross-metal mean

at or below the unconditioned baseline, which is the control that motivates adaptation rather than conditioning as the repair.

7 Conclusion

Metal identity is a first-class distribution shift for ligand-based activity models. On a deduplicated MetalCytoToxDB cohort, one shared model loses 0.06–0.24 ROC-AUC when it moves from metals it has seen to one it has not, and every metal incurs the loss. Few-shot metal adaptation — freezing the source model and fine-tuning only its activity head on a handful of labelled target-metal ligands — recovers most of it: the paired delta is positive at every K we test, reaches +0.0685 at K=80 with all 50 metal×seed cells positive, and recovers 74.9–98.5% of the within-metal ceiling. Matched-budget controls show the lift is the adaptation operator, not the extra labels or compute; a scope ablation shows that head-only adaptation is what keeps the low-K regime non-negative; a sampler ablation shows the method needs no careful support curation; and a fingerprint-plus-XGBoost replication shows the effect is not an artefact of the graph encoder.

The actionable reading is narrow but clear. A ligand-only QSAR model trained on one metal should not be deployed to rank candidates for a different metal without re-validation: the features that carry within-metal signal do not survive the metal change, and supplying the metal identity does not repair this. What does work is cheap — a few dozen labelled ligands from the target metal, a frozen encoder and a retrained head. We therefore recommend (i) few-shot adaptation to the target metal rather than a metal-conditioned zero-shot model, and (ii) reporting cross-metal leave-one-metal-out alongside the within-metal number whenever a dataset spans more than one metal. Cross-metal transfer deserves to be a reported benchmark axis, not an implicit assumption.

Data and code availability

All cohorts, configs and scripts are versioned in the accompanying repository. Every quantitative claim in this paper cites a JSON artifact under `metrics/by_metric/` or a run report under `molmetal/reports/`, listed at the corresponding table or figure.

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